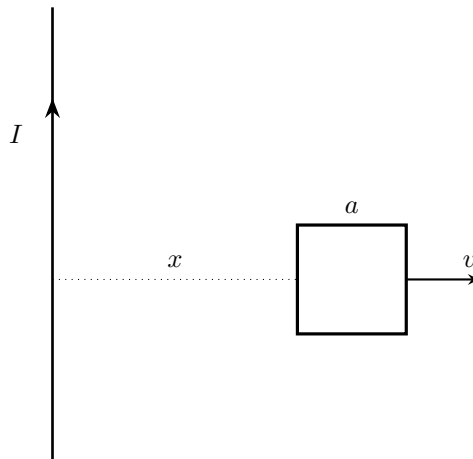


INPhO 2000

1. (a) Consider a charged water droplet of radius R and charge Q uniformly distributed over its surface. [10]
From energy considerations, obtain an expression for the minimum charge that permits the droplet to spontaneously split into (i) two droplets of equal size, (ii) three droplets of equal size. Show that both the processes are energetically allowed for a droplet of charge $Q = 5 \times 10^{-10}$ C and radius $R = 1$ mm. Will these processes actually occur if the dielectric breakdown of the surrounding medium takes place at field strength of 3×10^6 Vm $^{-1}$?
- (b) Monochromatic light of wavelength 500 nm and intensity 10^{-4} Wm $^{-2}$ falls normally on a sodium photocell of surface area 2 cm 2 . We adopt the classical wave picture of light and assume that the incident power is uniformly and continuously absorbed by the top 5 layers of the metal. Take the effective cross sectional area of each atom to be 10^{-20} m 2 . The work function for sodium is 2 eV. Estimate the time required for photoelectric emission (after the beam is incident on the photocell). Contrast your answer with the expectation in the photon picture of light. [10]
- (c) A container having nitrogen gas is maintained at a constant temperature by immersing it in a bath at 300 K. The container has a frictionless piston at one end and is connected to an evacuated bottle with non-conducting walls via a porous plug. The gas is allowed to flow through the plug while the pressure in the container is kept constant. Determine the temperature of the gas in the bottle when its pressure equals the pressure in the container. Neglect any heat conduction through the plug. Assume that nitrogen is an ideal gas. (Note: No heat is exchanged between the bath and the container throughout the process.) [10]
- (d) A collimated beam of light from a He-Ne laser ($\lambda = 632.8$ nm) is incident on a grating with 10 lines per mm. A converging lens system is placed in front of the grating and a screen is placed in the focal plane of the system. Draw the appropriate ray diagram and determine the focal length of the lens system if the diffraction maxima on the screen are 3.0 mm apart. [10]
- (e) Obtain an expression for the mutual inductance between a long straight wire and a square loop of side a as shown in the figure. [10]

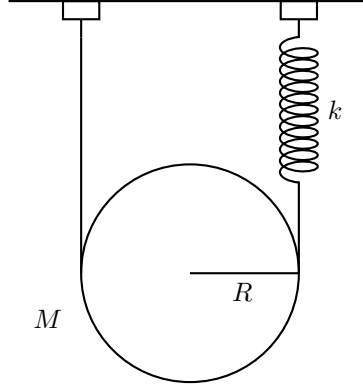


Evaluate the induced emf in the loop if the wire carries a current of 50 A and the loop has an instantaneous velocity $v = 10$ ms $^{-1}$ at the location $x = 0.2$ m, as shown. Take $a = 0.1$ m and assume that the loop has large resistance.

- (f) A string supports a disc of radius R and mass M as shown in the figure. One end of the string is connected to a rigid support while the other end is connected to a spring of force constant k . Show that the period of oscillation of the vertical motion of the disc in its plane, assuming that there is no slippage between the string and the disc, is given by

[10]

$$T = 2\pi\sqrt{\frac{3M}{8k}}$$



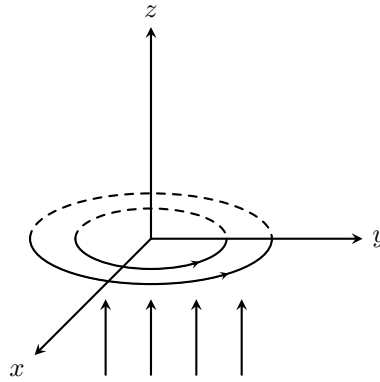
2. (a) A long uniform cylindrical beam of radius R consists of positively charged ions each of charge q , mass m and velocity v along the positive z -direction.
- (i) Show that an ion at the surface of the beam experiences a net outward force given by

[15]

$$F = \frac{Iq}{2\pi\epsilon_0 Rv} \left(1 - \frac{v^2}{c^2}\right)$$

where I is the beam current.

- (ii) The beam will diverge because of this outward force. We ignore this divergence here. Consider the beam to be incident on a region of magnetic field as shown:



The magnetic field in the region is given by

$$\vec{B} \begin{cases} rB_0\hat{\phi} & \text{for } 0 \leq z \leq \Delta z \\ 0 & \text{otherwise} \end{cases} \quad (B_0 \text{ is constant})$$

i.e. the magnetic field is confined within a region of small thickness Δz . Note r, ϕ and z are cylindrical co-ordinates ($x = r \cos \phi, y = r \sin \phi$). $\hat{\phi}$ is a small unit vector in the azimuthal direction: $\hat{r} \times \hat{\phi} = \hat{z}$. Show that the magnetic field acts as a converging lens for the ion beam, and obtain an expression for its 'focal length'. Calculate this focal length for a beam of protons with $v = 10^5 \text{ ms}^{-1}$, given that $B_0 = 0.1 \text{ Tm}^{-1}$ and $\Delta z = 1 \text{ cm}$.

- (b) (i) The total electromagnetic power output P of the Sun is 3.8×10^{26} W. Calculate the radiation pressure on the earth at normal incidence, assuming that the earth is a black body. Compare it with the normal atmospheric pressure. The mean orbital radius of the earth is 1.5×10^{11} m. [15]
- (ii) Kepler's III law for circular orbits of planets around the Sun is:

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

where R is the orbital radius, T the time period of revolution and M the solar mass. If the planet is a homogeneous sphere of density ρ and radius r , derive a modified form of the above law taking radiation pressure into account. Express the correction in terms of ρ , r and P .

- (iii) Consider a perfectly reflecting thin plane sheet of very light material located initially at rest (with respect to the Sun) at a point on the earth's orbit and facing the solar radiation normally. Determine the final speed attained by this 'solar sail' due to solar radiation. The surface mass density of the sheet is 3×10^{-4} kg m $^{-2}$. Ignore gravitational effects and assume that the sail faces the solar radiation normally throughout its flight.

3. A fusion reactor is designed to produce energy from nuclear fusion in a controlled way. We wish to investigate the conditions under which this is possible: [30]

- (a) A proposed fusion reactor is based on the deuterium-tritium fusion reaction (D-T reaction). Determine the net amount of energy (Q) in MeV released in a single D-T reaction using the following data on atomic masses:

$$\begin{aligned} m({}_1^2\text{H}) &= 2.014102 \text{ u} \\ m({}_1^3\text{H}) &= 3.016049 \text{ u} \\ m({}_2^4\text{He}) &= 4.002603 \text{ u} \\ m_n &= 1.008665 \text{ u} \end{aligned}$$

- (b) Neglecting initial kinetic energies and momenta, determine the kinetic energy carried by the product neutron in the D-T reaction.
- (c) Obtain an order of magnitude estimate of the temperature required to initiate the D-T reaction. You may take the radius of each nucleus to be 1.5×10^{-13} m.
- (d) At such high temperatures, the atoms are fully ionized and the fuel assembly is plasma. The hot plasma will lose energy by striking against the wall of the container and melt it, so a major technological challenge is to confine the plasma to the interior of the container. Suppose by some technique the plasma is confined for a short time. Show that the energy released per unit volume from fusion is given by

$$E_f = \frac{1}{4} n^2 \sigma v Q \tau$$

where n is the electron number density (densities of D and T ions are assumed to be equal), σ is the cross section for the fusion reaction, v is the speed of the ions and τ the confinement time. (σ stands for the probability of fusion reaction per unit time per unit flux of incident beam on a single target nucleus. Flux is the number of particles crossing a unit area per unit time.)

Note: The speed v above refers to the relative speed between the fusing nuclei. Further, since both the ions will have thermal distribution of velocities and σ may depend on v , the quantity σv above should, strictly, be a suitably defined average of σv over the velocity distributions. We ignore this complication here.

- (e) Show that the fusion reaction will produce a net energy gain if

$$n\tau > \frac{12kT}{\sigma v Q}$$

This is known as the *Lawson criterion*.

At the operating temperature 10^8 K, $\sigma v \sim 10^{-22}$ m 3 s $^{-1}$. Give an 'order of magnitude' argument for this numerical value of σv . Estimate the minimum ion density required for confinement time $\tau = 10^{-9}$ s for a fusion reactor to work.

- (f) Even when the plasma is confined, it can lose energy by other processes. One important source of energy loss is the energy radiated by the accelerated electrons and ions due to Coulomb interactions in the plasma. This process known as *bremsstrahlung* leads to power loss per unit volume given by the following formula for the D-T assembly:

$$P_{br} = \frac{8\pi^2 n^2 e^6 v_e}{3 (4\pi\epsilon_0)^3 m_e c^3 h}$$

where v_e is the thermal speed of the electrons. At the opening temperature $T = 10^8$ K, find the ratio of power released by fusion to the power loss by *bremsstrahlung*.